ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. ME ENGINEERING — SEMESTER 4

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Explain Fluid properties: Viscosity, Surface Tension, Capillarity. State Newton's Law of Viscosity. [14 Marks]

[OR]

Q.1 (Alt) State Pascal's Law. Calculate Total Hydrostatic Pressure and Center of Pressure on submerged surfaces. [14 Marks]

--- UNIT II ---

Q.2 Derive Continuity Equation in 3D Cartesian coordinates. Differentiate Streamline, Streakline, Pathline. [14 Marks]

[OR]

Q.2 (Alt) Derive Euler's Equation of Motion and Bernoulli's Theorem from first principles. State assumptions. [14 Marks]

--- UNIT III ---

Q.3 Explain Flow measuring devices: Venturimeter, Orifice meter, Pitot tube with discharge equations. [14 Marks]

[OR]

Q.3 (Alt) Derive Hagen-Poiseuille equation for laminar viscous flow through circular pipe. [14 Marks]

--- UNIT IV ---

Q.4 Explain Major and Minor losses in pipes. Derive Darcy-Weisbach equation for frictional loss. [14 Marks]

[OR]

Q.4 (Alt) Explain Boundary Layer theory: Laminar, Turbulent boundary layers, Separation, and control methods. [14 Marks]

--- UNIT V ---

Q.5 Explain Pelton Wheel Turbine: Velocity triangles, Work done, and Hydraulic efficiency. [14 Marks]

[OR]

Q.5 (Alt) Explain Francis Turbine and Centrifugal Pump: Priming, Cavitation, and Characteristic curves. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****